

DATA ENGINEERING COURSE

OOP in Python

Inheritance · Encapsulation · File Processing

Assignment

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Data Engineering Course

Student Instructions

Please read the following instructions carefully before attempting the assignment:

- Attempt all questions.
- Write clean, well-indented Python code following PEP 8 standards.
- Use meaningful variable and method names.
- Add comments where appropriate to explain your logic.
- Each class must be fully functional and testable.
- Test your code with multiple inputs, including edge cases.
- Submit your .py file with your name and roll number in the filename.

Classwork Assignment

Question 1: Encapsulation Practice

Create a class called **Student**.

Requirements:

- Create a private attribute called `__marks`.
- Create a method `set_marks()` to update marks.
- Create a method `get_marks()` to display marks.
- Marks should only be accepted if they are between 0 and 100.
- Display an appropriate message for invalid marks.

Question 2: Bank Account System

Create a class called **BankAccount**.

Requirements:

- Create a private attribute called `__balance`.
- Create the following methods:
 - `deposit()`
 - `withdraw()`
 - `get_balance()`
- Deposit amount must be greater than zero.
- Withdrawal amount cannot exceed the available balance.
- Display an appropriate message when funds are insufficient.

Day 19 — Homework Assignment

Question 1: Vehicle Management System

Create a parent class called **Vehicle**.

Requirements:

- Create a method called `start()` that displays a generic start message.

Create two child classes: **Car** and **Bike**.

Requirements for child classes:

- Inherit from **Vehicle**.
- Override the `start()` method in both child classes.
- Display vehicle-specific start messages.

Question 2: Product Inventory System

Create a class called **Product**.

Requirements:

- Create a private attribute called `__quantity`.
- Create methods:
 - `set_quantity()`
 - `get_quantity()`
- Quantity should never be negative.
- Display an appropriate message if an invalid quantity is entered.